# TripMind – UML Component Diagram

AI-Powered Travel Planner Web Application

**User**

- Enter travel preferences
- View generated itinerary
- Modify and regenerate

Interact via Web Browser

HTTP/HTTPS

## Web Front-end (React)

**Input Form**
(destination, dates, budget, travellers, interests)

**Display Results**
(itinerary, costs, travel info)

**Handle User Interaction**
(form validation, API calls, loading states, error messages)

REST API (JSON)

## Backend (Python / Flask)

### API Layer (Flask Routes)

- /api/generate (itinerary)
- /api/validate (optional)
- /api/attractions (optional)

### Travel Business Logic

- Process user input (dates, budget, interests)
- Retrieve relevant attraction data (if required)
- Apply filters and constraints (e.g. budget, days, interests)
- Generate structured itinerary

### LLM Interaction Component

- Build prompt with context (user input, constraints, history)
- Call LLM API (e.g. OpenAI / Claude / Gemini)
- Parse and structure LLM output (e.g. JSON)
- Handle errors and retries

### Data Access Layer

- Query / update data
- Return attractions and travel info

SQL Query

Query Result

API Request (HTTPS)

LLM Response (text / JSON)

## External LLM Services

- OpenAI (GPT)
- Anthropic (Claude)
- Google (Gemini)
- ••• (or other LLM providers)

## Database (SQLite)

- Travel data (attractions, cities, categories, etc.)
- User history (optional)
- Other supporting data

## Legend

- Component (UML)
- Interaction / data flow
- System boundary
- User / External Actor
- Front-end Component
- Backend Component
- Business Logic Component
- LLM Component
- External Service
- Database